Packet Tracer - Verify IPv4 and IPv6 Addressing

Addressing Table

Device	Interface	IP Address / Prefix		Default Gateway
R1	G0/0	10.10.1.97	255.255.255.224	N/A
R1	G0/0	2001:db8:1:1::1/64		N/A
R1	S0/0/1	10.10.1.6	255.255.255.252	N/A
R1	S0/0/1	2001:db8:1:1::1/64		N/A
R1	S0/0/1	fe80::1		N/A
R2	S0/0/0	10.10.1.5	255.255.255.252	N/A
R2	S0/0/0	2001:db8:1:2::1/64		N/A
R2	S0/0/1	10.10.1.9	255.255.255.252	N/A
R2	S0/0/1	2001:db8:1:3::1/64		N/A
R2	S0/0/1	fe80::2		N/A
R3	G0/0	10.10.1.17	255.255.255.240	N/A
R3	G0/0	2001:db8:1:4::1/64		N/A
R3	S0/0/1	10.10.1.10	255.255.255.252	N/A
R3	S0/0/1	2001:db8:1:3::2/64		N/A
R3	S0/0/1	fe80::3		N/A
PC1	NIC	Answer Area: 10.10.1.100	Answer Area: 255.255.255.224	Answer Area: 10.10.1.97
PC1	NIC	Answer Area: 2001:DB8:1:1::A		Answer Area: FE80::1
PC2	NIC	Answer Area: 10.10.1.17	Answer Area: 255.255.255.240	Answer Area: 10.10.1.17
PC2	NIC	Answer Area: 2001:DB8:1:4::A		Answer Area: FE80::3

Device	Interface	IP Address / Prefix		Default Gateway
R1	G0/0	10.10.1.97	255.255.255.224	N/A
R1	G0/0	2001:db8:1:1::1/64		N/A
R1	S0/0/1	10.10.1.6	255.255.255.252	N/A
R1	S0/0/1	2001:db8:1:1::1/64		N/A
R1	S0/0/1	fe80::1		N/A
R2	S0/0/0	10.10.1.5	255.255.255.252	N/A
R2	S0/0/0	2001:db8:1:2::1/64		N/A
R2	S0/0/1	10.10.1.9	255.255.255.252	N/A
R2	S0/0/1	2001:db8:1:3::1/64		N/A
R2	S0/0/1	fe80::2		N/A
R3	G0/0	10.10.1.17	255.255.255.240	N/A
R3	G0/0	2001:db8:1:4::1/64		N/A
R3	S0/0/1	10.10.1.10	255.255.255.252	N/A

R3	S0/0/1	2001:db8:1:3::2/64		N/A
R3	S0/0/1	fe80::3		N/A
PC1	NIC	10.10.1.100	255.255.255.224	10.10.1.97
PC1	NIC	2001:db8:1:1::a/64		fe80::1
PC2	NIC	10.10.1.20	255.255.255.240	10.10.1.17
PC2	NIC	2001:db8:1:4::a/64		fe80::3

Hide Answer

Objectives

Part 1: Complete the Addressing Table Documentation

Part 2: Test Connectivity Using Ping

Part 3: Discover the Path by Tracing the Route

Background

Dual-stack allows IPv4 and IPv6 to coexist on the same network. In this activity, you will investigate a dual-stack implementation including documenting the IPv4 and IPv6 configuration for end devices, testing connectivity for both IPv4 and IPv6 using **ping**, and tracing the path from end to end for IPv4 and IPv6.

Part 1: Complete the Addressing Table Documentation

Step 1: Use ipconfig to verify IPv4 addressing.

a. Click **PC1** and open the **Command Prompt.**

b. Enter the **ipconfig /all** command to collect the IPv4 information. Fill-in the **Addressing Table** with the IPv4 address, subnet mask, and default gateway.

c. Click **PC2** and open the **Command Prompt.**

d. Enter the **ipconfig /all** command to collect the IPv4 information. Fill-in the **Addressing Table** with the IPv4 address, subnet mask, and default gateway.

Step 2: Use ipv6config to verify IPv6 addressing.

a. On **PC1**, enter the **ipv6config /all** command to collect the IPv6 information. Fill-in the **Addressing Table** with the IPv6 address, subnet prefix, and default gateway.

b. On **PC2**, enter the **ipv6config /all** command to collect the IPv6 information. Fill-in the **Addressing Table** with the IPv6 address, subnet prefix, and default gateway.

Part 2: Test Connectivity Using Ping

Step 1: Use ping to verify IPv4 connectivity.

a. From **PC1**, ping the IPv4 address for **PC2**.

Was the result successful?

Área de respuesta

yes

Show Answer

b. From **PC2**, ping the IPv4 address for **PC1**.

Was the result successful?

Área de respuesta

yes

Show Answer

Step 2: Use ping to verify IPv6 connectivity.

a. From **PC1**, ping the IPv6 address for **PC2**.

Was the result successful?

Área de respuesta

yes

Show Answer

b. From **PC2**, ping the IPv6 address of **PC1**.

Was the result successful?

Área de respuesta

yes

Show Answer

Part 3: Discover the Path by Tracing the Route

Step 1: Use tracert to discover the IPv4 path.

a. From **PC1**, trace the route to **PC2**.

```
PC> tracert 10.10.1.20
```

What addresses were encountered along the path?

Área de respuesta

10.10.1.97
.5
.10

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With which interfaces are the four addresses associated

Área de respuesta

R2 s0/0/0
R3 s0/0/1

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b. From **PC2**, trace the route to **PC1**.

What addresses were encountered along the path?

Área de respuesta

.9
.6
.20

Show Answer

With which interfaces are the four addresses associated?

Área de respuesta

R2 s0/0/1
R1 s0/0/1
PC1 NIC

Show Answer

Step 2: Use tracert to discover the IPv6 path.

a. From **PC1**, trace the route to the IPv6 address for **PC2**.

```
PC> tracert 2001:db8:1:4::a
```

What addresses were encountered along the path?

Área de respuesta

2001:DB8:1:2::1
2001:DB8:1:3::2
2001:DB8:1:4::A

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With which interfaces are the four addresses associated?

Área de respuesta

R2 0/0/0 r2
R3 S0/0/1
PC2 NIC

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b. From **PC2**, trace the route to the IPv6 address for **PC1**.

What addresses were encountered along the path?

Área de respuesta

2001:DB8:1:4::1
2001:DB8:1:3::1
2001:DB8:1:2::2

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With which interfaces are the four addresses associated?

Área de respuesta

R2 s/0/0/1
R1 s/0/0/1
PC1 NIC

Show Answer

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